



Omnichannel Retail Behavior and Performance Analytics Dashboard

Analysis Period: March 2020 - March 2025

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Executive Summary

Project Overview

- ✓ This analysis was conducted as part of a business intelligence initiative to evaluate the **Omnichannel retail behavior and performance** of a fictional retail brand **OmniMart Pvt Ltd** over the period **March 2020 to March 2025**.
- ✓ The objective was to identify insights and opportunities across five key business areas - **Revenue, Products, Customers, Campaigns** and **Support**.

Business Objective

What OmniMart Aims to Achieve with the Power BI Dashboard ?



Accelerate Decision-Making

Deliver real-time, actionable insights to decision-makers across sales, marketing and support functions.



Unify Omnichannel Performance Visibility

Integrate sales, customer, campaign and support data from offline and online channels into one platform.



Enhance Customer Engagement & Retention

Track churn, campaign responses and SLA metric, CSS to improve satisfaction and loyalty.



Foster Accountability Through Role-Based Reporting

Provide tailored dashboards with access control for executives, managers, analysts and support teams.



Support Operational Efficiency and Data Trust

Replace manual reports with automated, validated dashboards powered by SQL and Power BI.

Data Source and Processing



About the Source

- ✓ The dataset was sourced from **Kaggle** website, a popular platform for hosting datasets.
- ✓ Kaggle hosts a wide range of community-contributed datasets, making it a valuable resource for analysis projects.
- ✓ This dataset contains fictional or sample retail data gathered from various sources within a retail business environment.



Description of the Dataset

(Click Hyperlink texts below to know more)

- ✓ Source: **Kaggle** - [🔗 Retail Customer & Transaction Dataset](#)
- ✓ Period Covered: March 2020 - March 2025
- ✓ Files Used:
 - [🔗 Customers](#)
 - [🔗 Transactions](#)
 - [🔗 Interactions](#)
 - [🔗 Campaigns](#)
 - [🔗 Customer reviews](#)
 - [🔗 Support tickets](#)



Data Preparation & Processing

- ✓ Verified file structures and column definitions in Excel.
- ✓ Checked quality: minor missing values (e.g., phone, discounts), assumed all amounts in **USD**.
- ✓ Imported datasets into **MySQL** for structured cleaning and transformation.



Data Cleaning

- ✓ Fixed missing values (dates, budget, ratings, product names).
- ✓ Removed duplicates (customer_id, emails).
- ✓ Validated formats (emails, phone numbers, names).
- ✓ Flagged orphan records; identified outliers in age, prices, satisfaction scores.
- ✓ Standardized store location (City/State).
- ✓ Standardized categorical values (e.g., store location formats).
- ✓ Recalculated and validated campaign KPIs.

Workflow and Approach



1. Requirement Gathering

- ✓ Created sample 'Business Request Email' format to identify reporting needs.



2. Business Requirement Document (BRD)

- ✓ Documented objectives, scope, KPIs, functional/non-functional requirements, success criteria, gap analysis.
- ✓ Created User Stories and Use Cases.



3. Process Mapping

- ✓ Designed current state and future state process flows using *Lucidchart*.



4. Wireframes & Layout Planning

- ✓ Built interactive dashboard wireframes in *Balsamiq* (summary, overview, revenue, products, customers, campaigns, support tickets).



5. Analysis Question Framework

- ✓ Developed 26 key analysis questions covering revenue, products, customers, campaigns and support operations to guide SQL development.



6. Data Preparation (see "Data Source" slide for details)

- ✓ Performed data cleaning, transformation and validation from multiple CSV files.



7. SQL Development

- ✓ Wrote 26 targeted SQL queries aligned with the analysis questions and KPIs.



8. Power BI Dashboard Development

- ✓ Imported cleaned datasets and applied data modelling best practices.
- ✓ Created DAX measures for KPIs and custom calculations.
- ✓ Designed interactive filters, slicers and drill-throughs.
- ✓ Applied conditional formatting and visual-level filters to highlight trends.



9. UAT (User Acceptance Testing)

- ✓ Created UAT plan, test cases, defect tracker, traceability matrix and sign-off.



10. Delivery

- ✓ Delivered final dashboard/report with supporting documentation.

Tools Used



Data Storage & Processing

- ✓ **MySQL** - Structured database creation, SQL-based cleaning, transformations and KPI mapping.
- ✓ **Excel** - Initial column verification, quick data quality checks and exploratory profiling.



Wireframing & Design

- ✓ **Balsamiq** - Created low-fidelity dashboard wireframes for Summary, Overview, Revenue, Products, Customers, Campaigns and Support pages.



Process Modelling

- ✓ **Lucidchart** - Designed *Current-state (As-is)* and *Future-state (To-be)* process flow diagrams to visualize reporting workflows.



Data Visualization & Reporting

- ✓ **Power BI** - Built interactive dashboards, applied DAX measures and developed page-level KPIs with slicers/filters for drill-downs.



- ✓ **MS Word** - Sample Business Request Email, Business Requirements Document (BRD)



- ✓ **MS Excel** - UAT plans, Test Cases, Defect Tracking.



- ✓ **MS PowerPoint** - Executive presentation of insights and recommendations.

Key Insights

Overview



Gross Revenue before discounts was **\$27.79M**, with total discounts reducing it by **\$0.24M**, resulting in a net revenue of **\$27.55M**



Net revenue generated is **\$27.55M**, with 'Furniture' category contributing the highest share of **\$6.1M**



'Google Nest' is the best-selling item with **1,359** units sold, while 'Sheets' leads in quality with an average rating of **5.0**



'California' state leads with the highest customer base of **873** individuals out of **5,000** total customers



'Search Engine Marketing' campaigns delivered the highest ROI of **\$162.64k**, with top performance seen in the 'Home Improvement' audience segment



'Shipping' was the most reported issue type, with **436** tickets raised, **379** of them were resolved, averaging **37.67** hours to close and a customer satisfaction score of **3.52**



Revenue Insights | Where the Money Flows



Overall

- ✓ Gross Revenue - **\$27.79M**
- ✓ Net Revenue - **\$27.55M**
- ✓ Discounts Applied - **\$240.77K**
- ✓ Discount Margin - **0.87%**, indicating a value-preserving pricing strategy rather than heavy discounting



Net Revenue Trends Over Time

- ✓ Strong YoY growth from 2020 to 2024, driven by post-COVID recovery and digital expansion
- ✓ Revenue growth peaked at **+231%** in 2021
- ✓ The **-87%** dip in 2025 is due to partial-year data (Jan-Mar) and is not reflective of performance decline



Seasonal Sales Highlights

- ✓ Peak sales months were November **\$3.71M** and December **\$3.79M**, highlighting strong holiday seasonality.



Top Payment Channel

- ✓ 'Credit Card' was the leading method
 - Revenue - **\$9.38M**
 - Transactions - **10,907**



Revenue by Channel & State

- ✓ Online Channel
 - Revenue - **\$13.97M**
 - 5× higher than any single physical region
- ✓ California
 - Revenue - **\$2.66M**
 - Top Performing Region



Top Product Categories by Revenue

- ✓ Furniture - **\$6.13M**
- ✓ Smartphones - **\$4.94M**
- ✓ Laptops - **\$3.94M**
- ✓ TVs - **\$3.25M**
- ✓ Kitchen Appliances - **\$2.91M**



Best-Selling Individual Products

- ✓ Bed Frame - **\$1.37M**
- ✓ Office Desk - **\$1.22M**
- ✓ Sofa - **\$1.21M**
- ✓ Bookshelf - **\$1.21M**
- ✓ Dining Table - **\$1.08M**

Products Insights | What People Bought



Overall

- ✓ Total Units Sold - **45,532**
- ✓ Average Product Rating - **4.0**



Avg Product Ratings

High Rated Products

- ✓ **5.0** Rating - iMac, Lenovo IdeaCentre, Lenovo Tab, Sheets
- ✓ **4.7** Rating - Baking Sheet, Cookware Set
- ✓ **4.5** Rating - Amazon Fire HD

Low Rated Products

- ✓ **3.0** Rating – Knife Set
- ✓ **2.5** Rating - Cast Iron Skillet
- ✓ **2.0** Rating - Logitech Mouse

Best-Selling Product Categories



- ✓ Smart Home Devices - **6,153** units
- ✓ Smartphones - **6,153** units
- ✓ Furniture - **5,575** units
- ✓ Home Decor - **5,349** units
- ✓ Kitchen Appliances - **5,255** units



Top 10 Products by Quantity Sold

- ✓ Google Nest - **1,359** units
- ✓ Xiaomi Mi 12 - **1,301** units
- ✓ Philips Hue - **1,272** units
- ✓ iPhone 13 - **1,257** units
- ✓ OnePlus 10 - **1,242** units
- ✓ Google Pixel 6 - **1,228** units
- ✓ Amazon Echo - **1,217** units
- ✓ Throw Pillows - **1,195** units
- ✓ Smart Thermostat - **1,186** units
- ✓ Bed Frame - **1,167** units



Revenue vs Discount

- ✓ Furniture and Smartphones yielded top revenue despite heavier discounts.
- ✓ Categories with higher net revenue tend to tolerate higher discount levels



First-Time Buyer Counts by Product Category

- ✓ Smartphones - **608** buyers
- ✓ Smart Home Devices - **600** buyers
- ✓ Kitchen Appliances - **529** buyers
- ✓ Furniture - **517** buyers

Customers Insights | Who They Are



Overall

- ✓ Total Customers - **5,000**
- ✓ Total Active Customers - **4,618**
- ✓ Customer Retention Rate - **27.86%**
- ✓ Customer Churn Rate (6 months) - **28.60%** (1,430 customers), mostly aged 36-50 and 18-25



Repeat Customers Count

- ✓ Repeat Purchase Rate - **83.94%**
- ✓ Preferred Channels:
 - Online - **2,056**
 - Both Online & In-store - **1,271**
 - In-store only - **775**
 - Unknown - **95**



Top 3 Customers by Revenue

- ✓ Tracey Patterson - California - **\$1.44M**
- ✓ Margaret Liu - Pennsylvania - **\$1.15M**
- ✓ Bethany Cowan - California - **\$1.06M**



Top States by Customers & Revenue

- ✓ California - **873** customers | **\$5.07M** revenue | Avg Order - **\$888**
- ✓ Texas - **644** customers | **\$3.57M** revenue | Avg Order - **\$863**
- ✓ Florida - **542** customers | **\$2.98M** revenue | Avg Order - **\$852**
- ✓ New York - **469** customers | **\$2.60M** revenue | Avg Order - **\$863**



Top Spend by Age & Gender

- ✓ Top Spending Segments
 - Female (36-50) - **\$5.23M**
 - Male (36-50) - **\$5.16M**
 - Male (26-35) - **\$4.27M**
 - Female (26-35) - **\$4.12M**
- ✓ Non-binary and other categories showed lower spend but consistent participation across age groups

Campaigns Insights | How We Reached Them



Overall

- ✓ Total ROI achieved - **\$296.86K**
- ✓ Overall conversion rate stands at **6.17%**



Peak Hours for Purchase Activity

- ✓ Purchases steadily increase starting from **5 AM**
- ✓ Peak hours are from **11 AM to 2 PM**, with the highest volume at **12 PM** with **829** purchases
- ✓ Activity declines after **4 PM**, with a small spike at **11 PM**



Top ROI-Efficient Target Segments

- ✓ "Home Improvement" leads with **\$59.09K ROI**
- ✓ "All Customers" generate **\$48.72K ROI**
- ✓ Other strong segments include New Customers, West Coast, Middle-aged 41-60 and Southern States generated more than **\$15k** each.



Campaign Performance by Type

- ✓ Search Engine Marketing performs best
 - ROI - **\$162.64K**
 - Conversion Rate - **10.45%**
 - Cost Per Click - **\$0.38**
 - Cost Per Acquisition - **\$3.67**
- ✓ In-Store Promotions show strong results
 - ROI - **\$48.63K**,
 - Conversion Rate - **29.05%**
- ✓ Underperforming campaigns
 - TV, Radio and Print Ads deliver low ROI under **1%** with very high acquisition costs
 - SMS and Influencer Marketing also have low ROI



Channel Engagement Insights

- ✓ Highest user interactions from Web with **55.7K** and Mobile App with **26.9K**
- ✓ Longest average session durations on Web at **91.08** minutes, followed by Mobile App at **82.49** minutes
- ✓ Mobile App drove the highest purchase actions at **13.6K**

Support Tickets Insights | How We Served Them



Overall

- ✓ **3,000** total tickets handled.
- ✓ Average Customer Satisfaction Score - **3.44**
- ✓ Ticket volume has grown steadily, peaking at **220+** in Jan 2025.
- ✓ **1 in 3 tickets** took more than **48** hours to resolve - area for improvement.
- ✓ Improve handling of “Account, Product Inquiry issues”.



Monthly Trends

- ✓ Steady growth in support volume over 5 years.
- ✓ No major drop in customer satisfaction despite rising ticket load.
- ✓ Recent months (2024 - 2025) handled **100+** tickets/month with stable CSS score.



Resolution Time Distribution

- ✓ **28.39%** of tickets resolved within 24 hours.
 - 0 - 12 hrs - **628** tickets
 - 12 - 24 hrs - **392** tickets
- ✓ **71.61%** of tickets took more than 24 hours to resolve.
 - 24 - 48 hrs - **971** tickets
 - 48+ hrs - **1,009** tickets



Ticket Status Overview

Status	Total Tickets	Avg Resolution Hrs	Avg CS Score
Resolved	2652	43.58	3.48
Pending	151	N/A	N/A
Escalated	86	N/A	N/A
Unknown	57	32.68	3.29
Closed without Resolution	54	124.00	1.89



Priority-Based Performance

Priority	Total Tickets	Avg Resolution Hrs	Avg CS Score
High	767	24.32	3.48
Medium	1328	35.80	3.55
Low	845	65.35	3.20
Unknown	60	45.48	3.83



Top Issue Categories

- ✓ Shipping: Tickets - 436 | Avg Res Hrs - 37.7 | Avg CSS - 3.52
- ✓ Account: Tickets - 430 | Avg Res Hrs - 51.1 | Avg CSS - 3.29
- ✓ Billing: Tickets - 424 | Avg Res Hrs - 33.6 | Avg CSS - 3.51
- ✓ Technical: Tickets - 419 | Avg Res Hrs - 33.15 | Avg CSS - 3.45

Recommendations



1. Revenue

- ✓ Focus marketing efforts on high-revenue categories like Furniture (\$8.13M), Smartphones (\$4.94M) and Laptops (\$3.95M).
- ✓ Capitalize on Nov (\$3.71M) and Dec (\$3.78M) surges with festive promotions, bundled deals and upselling.
- ✓ Optimize ads, personalized recommendations and loyalty rewards to further grow the \$13.98M Online revenue channel.
- ✓ Promote preferred payment methods like Credit Cards (\$9.38M across 10,097 transactions) to boost average order value and checkout speed.
- ✓ Reevaluate low-revenue categories under \$500K such as Tablets, Cookware, Small Kitchen Appliances, Bedding and Computer Accessories for promotional pushes.
- ✓ Develop location-specific offers to lift sales in Florida (\$1.19M), Colorado (\$1.22M) and similar low-performing states.



2. Product

- ✓ Increase stock levels by 20% for bestsellers (Bed Frame, Office Desk, Sofa - each \$1.2M+) before seasonal peaks.
- ✓ Expand promoting Smart Home Devices (6,153 units), Smartphones (6,153 units) and Furniture (5,575) with seasonal and event-based campaigns.
- ✓ Offer retention incentives to first-time buyers in Smartphones (608), Smart Home Devices (600) and Kitchen Appliances (529).
- ✓ Launch clearance or repositioning strategies for low-volume categories like Bedding, Computer Accessories, Desktop Computers and Tablets.
- ✓ Pair high-rated products such as iMac (5★) and Lenovo IdeaCentre (5★) with lower-selling items to boost sales.
- ✓ Highlight 4.5★+ rated products in ads to reinforce trust and improve conversion rate.

Recommendations



3. Customers

- ✓ Improve the 27.86% retention rate and reduce the 28.6% churn (1,430 in 6 months) by focusing on high-churn segments 26 - 35 (482 customers) and 50+ (170 customers) through tailored reactivation offers.
- ✓ Boost lifetime value from top-spending segments: Female 36 - 50 (\$5.23M), Male 36 - 50 (\$5.16M), and Male 26 - 35 (\$4.27M) with premium product bundles, early-bird offers, and loyalty perks.
- ✓ Leverage online repeat buyers (2,058 customers, 83.94% repeat purchase rate) through loyalty points, early access sales, and subscription models.
- ✓ Engage top spenders like Tracey Patterson (\$1.44M) and Margaret Liu (\$1.21M) with exclusive VIP programs to drive advocacy and repeat orders.
- ✓ Target high-value states California (\$5.08M), Texas (\$3.57M), Florida (\$2.99M) with geo-specific promotions while lifting lower-revenue states like Georgia (\$1.20M) and North Carolina (\$1.12M).



4. Campaigns

- ✓ Prioritize Search Engine Marketing (ROI - 32.31%) and Email Marketing (ROI - 15.92%) for future budget allocation, scaling high-ROI channels while reducing spend on low-ROI mediums like TV (0.16%) and Radio (0.34%).
- ✓ Optimize In-Store Promotions (Conversion Rate - 29.05%) by integrating QR codes and digital coupons to bridge offline-to-online conversions.
- ✓ Improve underperforming channels (e.g., Online Display ROI 1.13%, Social Media ROI 2.05%) by A/B testing creatives, tightening audience targeting, and leveraging retargeting strategies.
- ✓ Expand campaigns for Home Improvement (ROI - \$59K) while testing niche targeting for high-engagement segments like Technology Enthusiasts and Online Shoppers.
- ✓ Concentrate ad delivery during peak purchase hours 12 PM - 3 PM and 10 AM - 11 AM when purchase volume exceeds 750 orders/hour.

Recommendations



5. Support Tickets

- ✓ Ticket volume reached 3,000 in the latest period, with a sharp rise since mid-2023 and a peak of 220 in Jan 2025; capacity scaling is critical.
- ✓ Yearly trend shows a gradual rise in CS scores until late 2023, then fluctuating between 3.2 - 3.7, signalling stability but no significant improvement.
- ✓ Only 28.39% of tickets are resolved within 24 hrs, highlighting the need for faster triaging and improved first-contact resolution.
- ✓ High-priority tickets resolve in 24.32 hrs with a 3.48 CS score, but low-priority ones take 65.35 hrs and yield the lowest CS score (3.20), suggesting disproportionate delays.
- ✓ Shipping, Account Issues, and Billing each exceed 420 tickets; Account Issues have the longest resolution (51.13 hrs) and below-average satisfaction (3.29).
- ✓ Website Issues and Product Inquiries average 48+ hrs resolution, indicating system/process inefficiencies in these categories.
- ✓ “Closed without Resolution” cases take 124 hrs and deliver a poor 1.89 CS score; these should be minimized through escalation protocols.

Action Plan

Implement Targeted Revenue Strategies

Focus on high-performing states like California, Texas, and Florida while running pilot campaigns in underperforming regions to close the gap.

Optimize Product Mix

Increase inventory for top sellers and phase out or repackage low-margin SKUs, using seasonal demand trends to guide planning.



Enhance Customer Segmentation

Launch tailored marketing for high-value segments (e.g., Female 36-50, Male 26-35) while improving engagement for underrepresented demographics.

Refine Campaign Investments

Shift budget toward high-ROI channels such as Search Engine Marketing, Email, and In-Store Promotions, while testing new low-cost digital avenues.

Streamline Support Processes

Address high ticket volumes in categories like Shipping and Account Issues through automation, improved FAQs, and priority-based resolution workflows.

Appendix 1 - File Descriptions

1. Customers:

(Contains demographic, basic registration information about each customer.)

Rows - 5,000 | Columns - 12

Column Details:

- ✓ Customer ID (int or string): Unique identifier for each customer.
- ✓ Full Name (string): Full name of the customer.
- ✓ Age (int): Age of the customer.
- ✓ Gender (string): Gender of the customer (e.g., “Male”, “Female”, “Other”).
- ✓ Email (string): Email address.
- ✓ Phone (string): Phone number.
- ✓ Street Address (string): Street address.
- ✓ City (string): City of residence.
- ✓ State (string): State of residence.
- ✓ Zip Code (string): Zip code/postal code.
- ✓ Registration Date (date): Date the customer registered.
- ✓ Preferred Channel (string): Preferred communication channel (e.g., “Email”, “SMS”, “Phone”).

2. Transactions:

(Includes purchase information for products bought by customers.)

Rows - 32,295 | Columns - 10

Column Details:

- ✓ Transaction ID (int): Unique ID for each transaction.
- ✓ Customer ID (int or string): Links to customer id in customers.csv.
- ✓ Product Name (string): Name of the purchased product.
- ✓ Product Category (string): Category of the product.
- ✓ Quantity (int): Number of units purchased in this transaction.
- ✓ Price (float): Price per unit of the product.
- ✓ Transaction Date (date): Date when the transaction was completed.
- ✓ Store Location (string): Physical store location or “Online” if purchased via e-commerce.
- ✓ Payment Method (string): Payment method (e.g., “Credit Card”, “PayPal”, “Cash”).
- ✓ Discount Applied (float or int): Discount amount or percentage applied (if any).

Appendix 1 - File Descriptions

3. Interactions:



(Captures various interactions the customer has with the company's digital channels.)

Rows - 1,00,000 | Columns - 8

Column Details:

- ✓ Interaction ID (int): Unique ID for each interaction.
- ✓ Customer ID (int or string): Links to customer id in customers.csv.
- ✓ Channel (string): Channel of interaction (e.g., "Website", "Mobile App", "Social Media").
- ✓ Interaction Type (string): Type of interaction (e.g., "View Product", "Add to Cart", "Click Ad").
- ✓ Interaction Date (date): Date of the interaction.
- ✓ Duration (int or float): Duration of interaction in seconds or minutes.
- ✓ Page or Product (string): Page visited or product name related to the interaction.
- ✓ Session ID (string): Session identifier to track multiple actions within one session.

4. Campaigns:



(Details about marketing or advertising campaigns.)

Rows - 200 | Columns - 12

Column Details:

- ✓ Campaign ID (int): Unique campaign identifier.
- ✓ Campaign Name (string): Name or title of the campaign.
- ✓ Campaign Type (string): Type of campaign (e.g., "Email", "Social Media", "TV Ad").
- ✓ Start Date (date): Launch date of the campaign.
- ✓ End Date (date): Completion date of the campaign.
- ✓ Target Segment (string): Target audience or segment (e.g., "Loyal Customers", "New Registrations").
- ✓ Budget (float): Allocated budget for the campaign.
- ✓ Impressions (int): Number of impressions generated.
- ✓ Clicks (int): Number of clicks.
- ✓ Conversions (int): Number of desired conversions (e.g., sign-ups, purchases).
- ✓ Conversion rate (float): Percentage of impressions that resulted in conversions.
- ✓ ROI (float): Return on investment for the campaign.

Appendix 1 - File Descriptions

5. Customer Reviews:

(Collection of product reviews submitted by customers post-purchase.)

Rows - 1,000 | Columns - 10

Column Details:

- ✓ Review ID (int): Unique identifier for each review.
- ✓ Customer ID (int or string): Links to customer id in customers.csv.
- ✓ Product Name (string): Name of the product reviewed.
- ✓ Product Category (string): Category of the product reviewed.
- ✓ Full Name (string): Name of the customer (could be anonymized).
- ✓ Transaction Date (date): Date of the transaction related to this review.
- ✓ Review Date (date): Date the review was posted.
- ✓ Rating (int): Rating given by the customer (e.g., 1 to 5 stars).
- ✓ Review Title (string): Short title or summary of the review.
- ✓ Review Text (string): Full text of the customer's review.

6. Support Tickets:

(Tracks customer support interactions and resolution details.)

Rows - 3,000 | Columns - 10

Column Details:

- ✓ Ticket ID (int): Unique ID for each support ticket.
- ✓ Customer ID (int or string): Links to customer id in customers.csv.
- ✓ Issue Category (string): General category of the issue (e.g., "Billing", "Product Defect", "Returns").
- ✓ Priority (string): Priority level ("Low", "Medium", "High").
- ✓ Submission Date (date): Date the ticket was submitted.
- ✓ Resolution Date (date): Date the ticket was resolved (if applicable).
- ✓ Resolution Status (string): Status of the ticket ("Open", "Closed", "Pending", etc.).
- ✓ Resolution Time Hours (float): Time (in hours) taken to resolve the issue.
- ✓ Customer Satisfaction Score (int or float): Satisfaction rating after resolution (e.g., 1 - 5).
- ✓ Notes (string): Additional notes or remarks from the support agent.